Vision-Based Intravenous Therapy Monitoring: Fusing Fluid-Level, Drip-Rate, and Patient-State Cues from a Single Camera with Learned Time-to-Empty Prediction

Anonymous Author(s)

Anonymous Institution

Abstract. Intravenous (IV) therapy is monitored largely by manual checks, and delays in noticing an empty bag, a stalled drip, or a distressed patient can harm patients. Existing automated solutions rely on dedicated sensors such as load cells and microcontrollers, or on computer vision that addresses only one signal in isolation (fluid level, drip counting, or patient monitoring). We present a camera-only monitoring framework that removes weight-sensing hardware and combines three vision modules: a convolutional neural network for fluid-level estimation, a YOLO-based detector with tracking for drip-rate counting, and a pretrained facial-expression model for patient-distress estimation. A fusion layer merges these signals into a single tiered risk level, and a sequence model estimates the remaining infusion time from noisy vision-derived measurements instead of relying on a fixed volume-over-rate formula. We evaluate each module on held-out data and evaluate the fusion layer on real module outputs, reporting cross-setup generalization from public images to our own recordings. The fluid-level model reaches [TODO: X.X]% accuracy, drip counting reaches [TODO: X.X] mAP@0.5, and the time-to-empty model reaches an MAE of [TODO: X.X] minutes. The system runs on a commodity laptop and webcam, and we discuss its limitations, including the absence of clinical validation.

Keywords: Intravenous monitoring · Computer vision · Deep learning · Multi-signal fusion · Smart healthcare.

1 Introduction

Intravenous therapy is one of the most common hospital procedures, and nurses check infusion bags manually, often across many patients at once. Three failure modes are common: the bag runs empty before it is replaced, the drip rate drifts or stops (for example through blockage or infiltration), and the patient shows signs of distress that go unnoticed between rounds. Each is individually detectable, but they are typically monitored separately or not at all.

Automated IV monitors have generally taken one of two routes. Sensor-based designs measure bag weight with a load cell and amplifier, process readings on a

microcontroller, and push threshold alerts to a cloud dashboard. These require physical instrumentation of every stand, calibration, and per-bag wiring. Vision-based designs instead observe the bag or drip chamber with a camera. Prior vision work has shown that fluid level can be classified and drops counted from video with high accuracy under controlled conditions [TODO: cite the specific prior papers you read: level classification, drip counting with detection+tracking, and any deep-learning drip-monitoring work]. However, these efforts treat level, rate, and patient state as separate problems, evaluate on narrow, single-rig data, and issue alerts only once a fixed threshold is crossed.

This paper makes the following contributions.

- A camera-only IV monitoring pipeline that replaces weight-sensing hardware with learned fluid-level and drip-rate estimation.
- A fusion layer that combines IV status and patient state into a single tiered risk signal, so that combinations (for example a low bag together with visible distress) escalate differently from either cue alone.
- A learned time-to-empty estimator trained on noisy vision-derived level and rate histories, compared against the standard volume-over-rate baseline.
- An evaluation that reports per-module metrics, fusion behaviour on real module outputs, and generalization from public images to self-recorded footage.

To the best of our knowledge, among the works we surveyed, none combines all three signals in a single camera pipeline [TODO: re-verify with a fresh literature search before submitting; soften this sentence if any counterexample exists].

2 Related Work

Sensor-based IV monitoring. Load-cell systems estimate remaining volume from bag weight and forward readings through microcontrollers and IoT platforms, with threshold alerts through buzzers, displays, or mobile notifications [TODO: cite 2–4 IoT/load-cell IV monitoring papers]. They are accurate but hardware-dependent.

Vision-based fluid level. Image classification of IV bag fill level has been demonstrated with convolutional networks on small public datasets [TODO: cite the IEEE DataPort/ResNet-based paper and the public image datasets used]. Reported accuracy depends heavily on dataset size and how similar training and test conditions are.

Drip counting. Drop detection and tracking with object detectors and multi-object trackers has been applied to drip chambers [TODO: cite the YOLO+DeepSORT drip-counting paper and any other drip-monitoring work, including deep-learning drip monitoring with automatic flow cut-off]. These works focus on rate estimation and do not model the patient.

Patient monitoring. Facial-expression and pain estimation models are widely studied [TODO: cite 1–2 facial expression/pain recognition works and the datasets behind the pretrained model you chose], but are rarely coupled with infusion state.

[TODO: insert architecture diagram: camera → ROI split → M1 level / M2 drip / M3 distress → M4 fusion → M5 ETA and dashboard]

Fig. 1. Overview of the proposed camera-only monitoring pipeline.

Gap. The literature we reviewed addresses these signals separately, evaluates on narrow setups, and stops at threshold-based alerts. We target integration and robustness.

3 Proposed System

3.1 Overview

A single camera observes the IV bag, the drip chamber, and (optionally) the patient. Frames are cropped into regions of interest (ROIs) and passed to independent modules whose outputs are written to a shared state and consumed by the fusion layer, the time-to-empty model, and a dashboard (Fig. 1). Modules are independently callable so each can be retrained or replaced.

3.2 Fluid-Level Estimation (M1)

The bag ROI is classified into discrete fill levels ([TODO: state the class scheme used, e.g. empty / low / half / full, or percentage bins]) by a convolutional network. We fine-tune an ImageNet-pretrained backbone ([TODO: state backbone, e.g. MobileNetV2 or ResNet-18]) with a new classification head, using augmentation (brightness, contrast, small rotations and crops, blur) to reduce sensitivity to lighting and viewpoint. Transfer learning is used because public IV-bag image sets are small.

3.3 Drip-Rate Estimation (M2)

Drops in the drip-chamber ROI are detected by a YOLO-family detector and associated across frames by a tracker so that each falling drop is counted once. The rate r_t in drops per minute is computed over a sliding window. [TODO: state detector version, input resolution, tracker, and window length used]. Volume rate follows from the administration set's drop factor f (drops per mL): $Q_t = r_t/f$.

3.4 Patient-Distress Estimation (M3)

The patient ROI is passed to a pretrained facial-expression model whose outputs are mapped to a scalar distress score $d_t \in [0, 1]$ [TODO: name the pretrained model and the mapping from its classes to a distress score]. We do not train a new model on patient data. This module is included to demonstrate fusion and is not clinically validated.

Table 1. Illustrative fusion rules (thresholds are configurable).

Tier	Condition	Notification
Normal	level $> \tau_w$, rate within tolerance, $d_t < \tau_d$	none
Watch	one mild deviation	dashboard highlight
Alert	level $\leq \tau_w$, or stalled drip, or $d_t \geq \tau_d$	push notification
Critical	level $\leq \tau_c$, or Alert condition combined with distress	repeated notification

3.5 Fusion and Risk Scoring (M4)

Let ℓ_t be the estimated fill level, r_t the drip rate with prescribed target r^*, and d_t the distress score. The fusion layer maps (ℓ_t, r_t, d_t) to one of four tiers: *Normal*, *Watch*, *Alert*, *Critical*. Escalation is driven by (i) level below a warning or critical threshold, (ii) relative rate deviation $|r_t - r^*|/r^*$ above a tolerance or a stalled drip, and (iii) distress above a threshold, with combined conditions escalating one tier above their individual effect. Thresholds are configurable (Table 1). [TODO: replace with the exact rules and threshold values implemented in the code].

3.6 Learned Time-to-Empty Prediction (M5)

A common baseline estimates remaining time as $\hat{T}_t^{\text{base}} = V_t/Q_t$, where $V_t = \ell_t V_0$ is remaining volume. With vision-derived ℓ_t and r_t, measurement noise and quantized levels make this estimate unstable. We instead train a sequence model ([TODO: LSTM / GRU; state layers and window length]) that takes a window of $(\ell_{t-k:t}, r_{t-k:t})$ and outputs the remaining time. Training targets come from recordings where the true emptying time is known. We compare against $\hat{T}_t^{\text{base}}$ to test whether learning helps beyond arithmetic.

3.7 Dashboard

A lightweight web dashboard shows live level, rate, distress, tier, and predicted time-to-empty, and issues on-screen and push notifications. No physical buzzer, display, or microcontroller is required.

4 Experimental Setup

4.1 Data

Table 2 summarizes the data. Fluid-level training uses a public IV-bag image set of about 100 images [TODO: cite the exact dataset and its class scheme] plus [TODO: N] frames we recorded on our own rig. Drip-rate data are our own recordings of a drip chamber filmed at [TODO: N] distinct rates ([TODO: list rates]), under [TODO: describe lighting/background variation], with [TODO: N]

Table 2. Datasets used.

Module	Source	Size	Split
M1 level	public + self-recorded	[TODO: N] images	[TODO: train/val/test]
M2 drip	self-recorded	[TODO: N] clips / [TODO: N] drops	[TODO: train/val/test]
M3 distress	pretrained, no new data	–	–
M5 ETA	derived from M1/M2 runs	[TODO: N] runs	[TODO: train/val/test]

Table 3. Per-module results on held-out test data.

Module	Metric	Value
M1 level	accuracy / macro-F1	[TODO: XX.X] / [TODO: XX.X]
M2 drip	precision / recall / mAP@0.5	[TODO: XX.X] / [TODO: XX.X] / [TODO: XX.X]
M2 drip	drops/min absolute error	[TODO: X.X]
M5 ETA (learned)	MAE / RMSE (min)	[TODO: X.X] / [TODO: X.X]
M5 ETA (baseline V/Q)	MAE / RMSE (min)	[TODO: X.X] / [TODO: X.X]

manually labelled drop events. For M5, sequences are derived from [TODO: N] full emptying runs. No patient data were collected; M3 is evaluated only qualitatively on [TODO: describe what, e.g. volunteer team members with informed consent].

Splits are made so that test frames come from recording sessions not seen in training, to avoid near-duplicate leakage between adjacent video frames.

4.2 Training Details

[TODO: optimizer, learning rate, batch size, epochs, augmentation parameters, hardware, framework and version for each trained module.]

4.3 Metrics

M1 is evaluated with accuracy, macro-F1, and a confusion matrix. M2 uses precision, recall, and mAP@0.5 for detection, plus the absolute error of counted drops per minute against manual counts. M5 uses MAE and RMSE in minutes and is compared with the volume-over-rate baseline. M4 is evaluated on real module outputs by alert precision, recall, and alert lead time against manually annotated events. To test generalization we train M1 on public images only and test on our recordings, and compare with training on the combined set.

5 Results and Discussion

[TODO: Fill every number from real runs only. Do not estimate or round up.]

5.1 Fluid Level

[TODO: Report Table 3 M1 numbers, include the confusion matrix figure, and discuss which adjacent levels are confused. Report the public-only vs. combined training comparison to quantify the generalization gap.]

5.2 Drip Rate

[TODO: Report detection metrics and rate error, discuss failure cases such as fast drips, motion blur, glare, and drops merging into a stream.]

5.3 Time-to-Empty

[TODO: Compare learned vs. baseline. If the learned model does not beat the baseline, say so and report the conditions where it does or does not help, for example near empty when level quantization dominates.]

5.4 Fusion Behaviour

[TODO: Show representative timelines where combined cues escalate earlier or differently than independent alerts, and report alert precision/recall/lead time on annotated events. If fusion was only run on limited data, state that plainly.]

6 Limitations

This is a prototype. The fluid-level model is trained on a small image set and may not generalize across bag brands, fill materials, camera angles, and lighting beyond those tested. Drip counting depends on an unobstructed, well-lit view of the drip chamber. The distress module uses a pretrained expression model that has not been validated on clinical populations, and expression does not reliably indicate clinical deterioration. Fusion thresholds are hand-set and were not tuned on patient data. The system has undergone no clinical evaluation and must not be used for patient care. Continuous camera monitoring in wards also raises privacy and consent concerns that require institutional review before any deployment.

7 Conclusion and Future Work

We presented a camera-only framework that estimates IV fluid level and drip rate, estimates patient distress, fuses them into tiered alerts, and predicts time-to-empty with a learned model, without weight-sensing hardware. Future work includes larger and more varied datasets across bag types and settings, learned rather than hand-set fusion, on-device deployment on low-cost edge hardware, and clinically supervised validation.

References

1. Redmon, J., Divvala, S., Girshick, R., Farhadi, A.: You Only Look Once: Unified, Real-Time Object Detection. In: Proc. IEEE CVPR, pp. 779–788 (2016)
2. He, K., Zhang, X., Ren, S., Sun, J.: Deep Residual Learning for Image Recognition. In: Proc. IEEE CVPR, pp. 770–778 (2016)
3. Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., Chen, L.-C.: MobileNetV2: Inverted Residuals and Linear Bottlenecks. In: Proc. IEEE CVPR, pp. 4510–4520 (2018)
4. Wojke, N., Bewley, A., Paulus, D.: Simple Online and Realtime Tracking with a Deep Association Metric. In: Proc. IEEE ICIP, pp. 3645–3649 (2017)
5. Hochreiter, S., Schmidhuber, J.: Long Short-Term Memory. Neural Computation 9(8), 1735–1780 (1997)